Mapping Reionization-Era Ionized Bubbles by Recovering Lyman-α Information from Low-Resolution JWST and Roman Spectroscopy

Science/Technical/Management Section

1 Science Goals and Objectives

The Epoch of Reionization (EoR) — the last major phase transition of the Universe, during which the first luminous sources ionized the neutral intergalactic medium (IGM) — remains one of the frontier problems of galaxy formation and cosmology [1,2]. While the broad timing of reionization is now bracketed ($z \sim 5.5$–10; [3,4]), its *morphology* is not: we do not yet know the characteristic sizes of ionized bubbles at a given epoch, how they grew and merged, or which galaxy populations drove that growth. The ionized bubble size distribution (BSD) as a function of redshift is the key observable that discriminates between reionization scenarios — early versus late, gradual versus rapid, and driven by faint numerous galaxies versus rare luminous systems [5,6,7].

The Lyα emission line of $z \gtrsim 7$ galaxies is the most direct available probe of this morphology. A Lyα photon escaping a galaxy embedded in a small ionized bubble encounters nearby neutral hydrogen and is strongly attenuated by the damping wing; a photon emitted inside a large bubble redshifts safely out of resonance before reaching the neutral IGM and survives [8,9]. Consequently the visibility, equivalent width, and velocity profile of Lyα encode the size of the local ionized region [10,11]. JWST has transformed this field by detecting Lyα from galaxies deep into the EoR, including surprisingly strong emitters at $z = 7$–11 interpreted as sitting inside early large bubbles [12,13,14]. Yet current constraints rest on small samples of individually modeled objects. Measuring the BSD — a population-level statistic — requires extracting IGM transmission information from *hundreds* of spectra, and there lies the obstacle: statistically large EoR samples exist almost exclusively at low spectral resolution, where the diagnostic information in the line is smeared away. The JWST/NIRSpec prism ($R \sim 30$–300) dominates existing $z > 7$ spectroscopy, and the Roman Space Telescope grism ($R = 461\lambda/\mu$m over 1–1.93 μm) will soon deliver the widest-area EoR spectroscopy ever taken — both at resolutions where Lyα velocity structure is unresolved.

This proposal will break that resolution barrier with a validated, physics-informed deep-learning framework, and use it to deliver the first statistical measurement of the ionized bubble size distribution at $z = 7$–9 from Lyα observables. Recent work [15] demonstrated a physics-informed super-resolution (SR) framework that enhances real JWST/NIRSpec prism spectra by a factor of ~10 in resolving power ($R \sim 100 \rightarrow R \sim 1000$), trained on ~1,200 paired prism–grating observations from the public JADES survey; the framework deblends closely spaced emission features (e.g., the [O III]$\lambda\lambda$4959,5007 doublet and Hβ), improves emission-line S/N by factors of several, and returns calibrated, wavelength-dependent uncertainties. The proposed research builds on this foundation through three objectives:

- **Objective 1 — Validate Lyα information recovery from low-resolution spectra.** Quantify, using held-out JADES prism–grating pairs and simulation-based falsification tests, which Lyα diagnostics (equivalent width, line centroid relative to systemic redshift, and damping-wing continuum shape) are recoverable from prism data, and compare super-resolve-then-infer against direct low-resolution inference (§3.1).
- **Objective 2 — Measure the bubble size distribution at $z = 7$–9.** Couple the validated spectral inference to a neural posterior model trained on radiative-transfer (RT) simulations

of the reionizing IGM, apply it to the public JADES $z = 7$–9 prism sample (>100 galaxies), and infer the population-level BSD with a hierarchical framework that uses detections, non-detections, and upper limits (§3.2).

- **Objective 3 — Make the measurement Roman-ready.** Adapt the framework to Roman grism spectroscopy, where Lyα enters the bandpass at $z \gtrsim 7.2$, using public Roman grism simulation products and an already-working end-to-end extraction pipeline, so that the BSD measurement can be extended to the rare, luminous galaxies tracing the largest bubbles as soon as Roman survey data become public (§3.3).

If successful, this program will convert the largest existing and forthcoming EoR spectroscopic samples — currently used mainly for redshift confirmation — into a quantitative map of reionization's morphology, providing a direct observational test of whether ionized bubbles at $z \sim 8$ follow the size distributions predicted by galaxy-driven reionization models [5,6] or require rarer, more luminous drivers [14,16]. All data used are from NASA missions: public JWST archival spectroscopy and Roman preparation products.

2 Scientific Background

2.1 Ionized bubbles and Lyα observables

During reionization, ionized regions grow around biased sources, percolate, and merge; analytic models and simulations predict a characteristic bubble scale growing from $\lesssim$1 comoving Mpc when the IGM is mostly neutral to tens of cMpc as reionization completes [5,6,7]. For a galaxy at systemic redshift $z_{\rm sys}$ inside a bubble of proper radius R_b, the Lyα damping-wing optical depth at line center scales inversely with the distance the photon travels before encountering neutral gas; transmission rises steeply with R_b up to ~1 pMpc [8,10]. Three observables carry this signal at low resolution: (i) the Lyα *equivalent width* distribution, whose evolution at $z > 6$ is the classic neutral-fraction probe [11,17]; (ii) the *velocity offset* of the transmitted line peak from systemic — larger offsets survive smaller bubbles, so the offset distribution maps bubble sizes when systemic redshifts are known from other lines [10,18]; and (iii) the *damping-wing imprint* on the continuum redward of the break [9,19]. JWST spectroscopy has produced both striking individual cases — strong Lyα at $z = 8$–11 implying early $\gtrsim$1–3 pMpc bubbles [12,13,14] — and growing statistical samples [20,21]. What is missing is a method that extracts profile-level information from the low-resolution spectra that constitute the bulk of the data, and a forward-modeling framework that turns a heterogeneous population of detections and limits into a BSD measurement. That is precisely what this proposal provides.

2.2 The low-resolution bottleneck and validated super-resolution

At the NIRSpec prism resolution near 1 μm ($R \sim 50$), one resolution element spans several thousand km s^{-1}: Lyα velocity offsets (typically 100–500 km s^{-1}) and line asymmetries are unresolved, and the line blends into the continuum break. Medium-resolution gratings resolve these diagnostics but exist for only a small fraction of EoR targets. The SR framework of [15] was built to bridge exactly this gap: a three-stage architecture (a conservative convolutional SR stage with per-pixel uncertainty; a redshift-inference head; and a physics-informed refinement stage in which ~80 known emission features are modeled as attention-gated parametric line profiles added to a smooth continuum correction) trained end-to-end on *real* paired prism/grating JADES observations rather

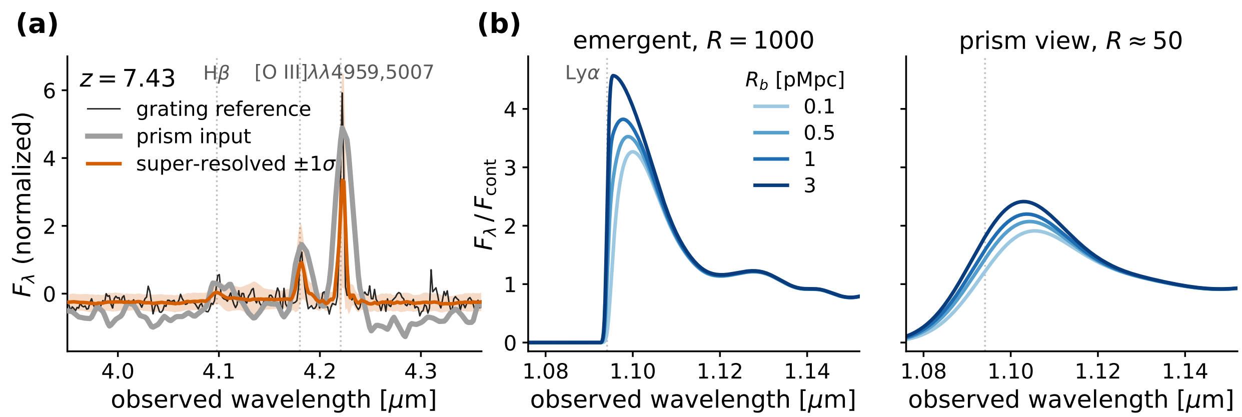

Figure 1: **(a)** A JADES galaxy at $z = 7.43$: the NIRSpec prism spectrum (grey), the medium-resolution grating reference (black), and the super-resolved output of the framework [15] with 1σ uncertainty (color). The prism blends Hβ and the [O III] doublet into a single broad feature; the super-resolved spectrum recovers all three lines at grating resolution. **(b)** A fixed intrinsic Lyα profile (median of JADES $z \simeq 4$–5.5 Lyα-emitter templates) transmitted through ionized bubbles of $R_b = 0.1$–3 pMpc at $z = 8$ (Miralda-Escudé damping wing, $\bar{x}_{\rm HI} = 0.5$ outside the bubble), shown at $R = 1000$ (left) and convolved to prism resolution (right): the bubble-size signature that is unresolved in the prism data is the information this program recovers and exploits.

than idealized simulations. On held-out test data the framework achieves noise-limited residuals over most of the spectral range and recovers redshift information approaching that of the high-resolution reference spectra [15].

Two properties make this framework suitable for trustworthy *science* — not just visually sharper spectra. First, it produces calibrated wavelength-dependent uncertainties, enabling propagation into downstream inference. Second, and critically, an accompanying falsification methodology (a "prior-dominance" audit) measures whether the model reads the data or recites its training prior: known perturbations are injected into the underlying high-resolution truth, forward-modeled into the low-resolution input, and the recovered response is measured as an exponent r ($r = 1$: fully data-driven; $r = 0$: pure prior). This audit, developed in the Roman-adaptation work described in §3.3, exposed prior-dominated behavior in an unaugmented model ($r \simeq 0.35$) and directly motivated the physics-based augmentation strategy that mitigates it. This built-in capacity for self-falsification addresses the central and legitimate skepticism about ML-reconstructed spectra, and is carried through every task below.

3 Technical Approach and Methodology

3.1 Task 1: Validated Lyα recovery from prism spectra (Year 1)

T1.1 — Lyα training and validation set. From the 1,187 paired prism–grating observations already assembled from the public JADES data releases [22], the subset with coverage of Lyα at $z > 5.5$ will be identified with the existing ingestion pipeline. Because the IGM is largely transparent at $z \simeq 4$–5.5, several hundred additional JADES spectra at those redshifts provide intrinsic-profile templates; a mock-generation pipeline (already prototyped) multiplies these templates by simulated IGM transmission curves and convolves to prism resolution, yielding a controlled test bed where the true transmission is known.

T1.2 — Fine-tune and audit the SR framework on the Lyα region. The pretrained framework of [15] will be fine-tuned with a loss emphasizing the Lyα line and adjacent continuum, including the resonant-line asymmetry. The prior-dominance audit will be extended to Lyα: transmission curves perturbed off the training manifold (systematically larger/smaller bubbles, damping-wing shapes from different $x_{\rm HI}$) are forward-modeled to prism resolution and the response exponent r measured per diagnostic. Acceptance criterion: $r \gtrsim 0.7$ for equivalent width and centroid; diagnostics failing the audit are excluded from downstream inference rather than patched.

T1.3 — The decisive comparison: SR-then-infer vs. direct inference. Information not present in the prism data cannot be created by SR; its value must be demonstrated, not assumed. On identical mocks, two chains will be compared end-to-end: (a) direct neural inference of transmission parameters from the prism spectrum, and (b) inference from the super-resolved spectrum with propagated uncertainties. The comparison — per-diagnostic bias, scatter, and calibration — fixes the architecture for Task 2 and is itself a publishable methods result (planned publication 1). Either outcome advances the science: if SR adds recoverable information (as the deblending results of [15] indicate), the gain is quantified; if not, the direct chain proceeds with the same validation machinery. *This task therefore cannot fail its own premise.*

3.2 Task 2: The bubble size distribution at z = 7–9 (Years 1–2)

T2.1 — Simulation training set. Per-sightline Lyα transmission curves with known local bubble sizes will be drawn from RT simulations of patchy reionization spanning $z \simeq$ 6–9 and a range of global neutral fractions. Three sources ensure robustness against simulation choice: (i) a suite of RT post-processed hydrodynamic simulations already in hand (obtained via private consultation), providing 10^4 sightlines per snapshot with matched ionization cubes from which multi-ray mean-free-path bubble labels are computed (pipeline already built and tested); (ii) the public THESAN Lyα transmission catalogs [23]; and (iii) the public SPHINX$_{20}$ data release [24]. Cross-simulation training/testing quantifies model dependence of the labels — reported as a systematic error component, not hidden.

T2.2 — Per-galaxy posterior inference. A neural posterior estimator (simulation-based inference; [25]) will map the recovered Lyα observables — with their calibrated uncertainties — to a posterior over local bubble size R_b and line-of-sight transmission, conditioned on galaxy properties (UV magnitude, systemic redshift quality). Mock-recovery experiments establish coverage (posterior calibration) before any application to data.

T2.3 — Population-level BSD. A hierarchical Bayesian layer combines per-galaxy posteriors — including non-detections and upper limits, which carry much of the information at $z > 7$ — into the BSD at $z \simeq$ 7, 8, and 9, marginalizing over intrinsic EW distribution parameters [11]. The result will be confronted with predictions of galaxy-driven semi-numeric models [6,7] and the RT suites of T2.1.

T2.4 — Application sample. The primary sample is the public JADES prism sample: 108 galaxies at z = 7–8 are already extracted with the existing pipeline, to be extended to z = 8–9 from the same public releases and supplemented by other public NIRSpec programs whose prism spectra cover Lyα at $z > 7$ — CEERS, UNCOVER, and RUBIES, all with team-released reduced spectra (via MAST and the DAWN JWST Archive). Systemic redshifts come from strong rest-optical lines in the same spectra ([O III], Hβ), for which the SR framework was originally validated [15].

Planned publication 2 (flagship): the first statistical BSD at $z = 7$–9 from Lyα observables.

3.3 Task 3: Extension to Roman grism spectroscopy (Years 2–3)

Roman's wide-area grism spectroscopy will observe of order a thousand square degrees at $R = 461\lambda/\mu$m over 1–1.93 μm [26,27]; Lyα enters this bandpass at $z \gtrsim 7.2$. Roman will thus capture the bright ($M_{\rm UV} \lesssim -22$), rare galaxies that JWST's deep pencil beams miss — exactly the population expected to trace the largest ionized bubbles [14,16] — over volumes two orders of magnitude beyond existing surveys.

T3.1 — Simulation-based grism pipeline (complete/in hand). An end-to-end pipeline operating on the public Roman grism simulation products [28] (hosted at IRSA) has already been developed as groundwork: it prepares the simulated 2D slitless frames, builds full-field contamination models, performs optimal 1D extraction with a wavelength solution validated to sub-pixel accuracy (median error −3 Å on emission-line knots), and has produced >17,000 low-/high-resolution training pairs, with ~250,000 expected from the full 15-field products. The SR stage retrained on these data recovers emission lines at 80–95% of true amplitude for line S/N $\gtrsim 5$.

T3.2 — High-z injection and Lyα recovery. The public simulation products contain $0 < z < 3$ galaxies only; EoR sources will therefore be *injected*: mock Lyα lines and damping-wing continua from T2.1, plus realistic interlopers, dispersed into the 2D frames through the validated pipeline, then recovered end-to-end. This yields Roman-specific completeness, purity, and R_b-recovery functions versus magnitude and line flux — the quantitative forecast for Roman's EoR bubble census (planned publication 3).

T3.3 — Deployment path and data risk statement. This project does *not* depend on unreleased data: Objectives 1–2 use public JWST archives, and Objective 3's deliverables are validated on public simulation products. Real Roman wide-area spectroscopy, expected within the award period following launch (~2027), will be used as an opportunity as soon as it is public: the T3.2 machinery applies without modification, and cross-instrument overlap samples will anchor sim-to-real domain adaptation. If Roman data are delayed beyond the award period, the contingency is expansion of Task 2 to the full public JWST archive; Objectives 1–2 and all three publications remain achievable.

3.4 Anticipated setbacks and mitigation

- *SR adds little for Lyα specifically* → T1.3 is designed as a decision experiment; the direct-inference chain uses the same validation machinery (methods paper either way).
- *Simulation dependence of bubble labels* → three independent simulation suites (T2.1); cross-training systematic budget; prior-dominance audit per diagnostic.
- *Small Lyα-emitter samples at $z > 8$* → the hierarchical BSD framework (T2.3) is built for censored data; non-detections constrain the small-bubble end.
- *Roman schedule slip* → explicit JWST-only contingency (T3.3).

3.5 Computing resources

Model training is modest by ML standards (~2.5M-parameter networks; datasets $\lesssim$10 GB): all tasks are executable on a dedicated research-group GPU system (NVIDIA RTX 5090 class) supplemented by the institutional GPU cluster, both described in the Expertise & Resources document. No NASA High-End Computing request is included.

4 Relevance to NASA and the Astrophysics Division

This investigation directly addresses the Astrophysics Division goals to "probe the origin and destiny of our universe" and "explore the origin and evolution of the galaxies," and is relevant to the Astrophysics Research program. It is built exclusively on NASA mission data and preparation products: public JWST/NIRSpec archival spectroscopy from MAST (Objectives 1–2) and Roman grism simulation products from IRSA in direct preparation for Roman's core community surveys (Objective 3). Beyond the specific EoR science, the validated spectral-inference methodology — with built-in falsification audits and calibrated uncertainties — is directly transferable to Roman's survey-scale emission-line science and to spectroscopy with future missions such as the Habitable Worlds Observatory. All software, trained models, and catalogs will be publicly released (see OSDMP).

5 Work Plan, Schedule, and Milestones

Year 1	T1.1–T1.3 complete; Lyα audit results; SR-vs-direct decision; T2.1 multi-simulation training set assembled. *Milestone:* methods paper submitted (publication 1). Conference presentation (AAS).
Year 2	T2.2–T2.4: posterior calibration, hierarchical BSD from JADES + public archive at z = 7–9. *Milestone:* flagship BSD paper submitted (publication 2). T3.2 injection framework running.
Year 3	T3.2–T3.3: Roman forecast paper (publication 3); application to early public Roman data if available; public release of code, trained models, and catalogs; thesis completion.

Table of Personnel and Work Effort (anonymized; the identical table with names appears in the Expertise & Resources document):

Role	Contribution	Effort
FI	All analysis, pipeline development, publications	1.0 FTE
PI	Thesis supervision, survey/spectroscopy guidance	no cost requested
Mentor 1	ML methodology and spectroscopy co-mentoring	no cost requested

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Open Science and Data Management Plan

(a) Data types, volumes, formats, and standards. This project uses only publicly available input data: JWST/NIRSpec spectra from public JADES data releases and other public programs, retrieved from MAST; and public Roman grism simulation products hosted at IRSA. Radiative-transfer simulation sightlines are obtained from the public THESAN and SPHINX releases and from a simulation suite available to the team (obtained via private consultation); only derived, shareable products of the latter (transmission curves and bubble-size labels used for training) will be redistributed, with permission. Generated products comprise: (i) curated training datasets (HDF5/npz; $\lesssim$10 GB total); (ii) trained model weights with configuration files (<1 GB); (iii) per-galaxy value-added catalogs (Lyα diagnostics, posteriors over bubble size; FITS and ECSV following IVOA-compliant column metadata; <1 GB); and (iv) all analysis software (Python). Formats follow community standards (FITS, HDF5, ECSV; PyTorch checkpoint format for weights).

(b) Repositories. Software will be developed openly in a public git repository from day one and archived at Zenodo (with DOI) at each release. Trained models and training datasets will be deposited on Zenodo and mirrored on a public model hub. Value-added catalogs will be published as machine-readable supplements to the associated journal articles, deposited on Zenodo, and contributed to MAST as a High-Level Science Product if accepted. Interim storage uses the research group's dedicated GPU server, which is sufficient for the training sets and model checkpoints involved; multi-TB project storage on the institutional high-performance computing center is available should additional capacity be required.

(c) Release timeline. Each dataset, model, and catalog is released no later than publication of its associated paper (planned: methods, Year 1; BSD, Year 2; Roman forecast, Year 3). As-accepted manuscripts are posted to arXiv at acceptance. Any scientifically useful product not covered by a publication is released by the end of the award.

(d) Archiving responsibility and experience. The FI performs all curation and archiving, and has prior experience releasing research software, trained checkpoints with model cards, and documented pipelines through public repositories and model hubs (phrased here without identifying links; see the Expertise & Resources document). The PI's group maintains long-term stewardship of the group's public data products.

(e) Software development and release. Software is developed under version control with continuous-integration tests and documentation, released under a permissive open-source license (BSD-3/MIT), with pinned environments (lock files) and end-to-end examples sufficient to reproduce every figure and catalog in the publications. Persistent identifiers: all team members maintain ORCIDs linked in NSPIRES; datasets and software carry Zenodo DOIs cited in the papers.

Mentoring Plan

This plan was prepared jointly by the FI, the PI, and one co-mentor (Mentor 1: machine-learning methodology and space-based spectroscopy). It complements the research plan by defining how the FI acquires the scientific breadth, technical depth, and professional skills to complete the proposed program and transition to an independent research career.

1. Mentoring structure and cadence. The FI meets the PI weekly (research direction, progress against milestones) and Mentor 1 weekly by video, supplemented by in-person working sessions at Mentor 1's institution approximately every two weeks (model design, validation methodology, survey spectroscopy). Guidance on IGM physics and the reionization simulations is available through established expertise at the FI's institution, including on the FI's thesis committee. A joint all-mentors meeting is held quarterly to review milestones from the Work Plan (§6 of the S/T/M) and adjust scope; written quarterly goals are agreed and revisited. The FI's institutional thesis committee provides an independent annual assessment.

2. Scientific and technical development. *Year 1:* depth in reionization theory and Lyα radiative transfer (directed reading guided by the PI and institutional experts; graduate IGM coursework as needed); statistical inference (simulation-based inference, hierarchical modeling); completion of the validation-methodology skill set with Mentor 1. *Year 2:* population-level statistical analysis; scientific writing leadership on the flagship paper; peer-review apprenticeship (co-refereeing with the PI where journals permit). *Year 3:* survey-scale data engineering for Roman-ready deployment; supervision experience by co-mentoring a junior student on a bounded sub-project; thesis completion.

3. Professional development and community integration. The FI presents annually at a major national meeting (e.g., AAS) and at one topical workshop per year (EoR/Lyα or astro-ML), with mentors coaching talk preparation. The FI's regular in-person working sessions at Mentor 1's institution (approximately biweekly; §1) embed the FI in that institution's survey-science community and build collaboration networks beyond the immediate mentoring team. The FI participates in the relevant Roman science community activities to prepare for survey data, and in journal clubs and group meetings across the mentoring team. Travel costs are budgeted in the E&R document.

4. Career planning. Semi-annual career conversations with the PI cover trajectory, fellowship and postdoctoral applications (application cycle prepared in Year 3 with mentor review of materials), and building a recognizable independent research identity at the intersection of EoR science and trustworthy ML. The mentoring team commits to timely recommendation letters and to nominating the FI for early-career opportunities (conference talks, science-team memberships).

5. Assessment. Progress is assessed against the milestone table in the S/T/M: submission of publications 1–3, public releases of code/models/catalogs, and conference presentations. If a milestone slips by more than one quarter, the mentoring team and FI jointly re-scope using the mitigation options identified in the research plan (§3.4 of the S/T/M). The plan itself is revisited annually and updated by mutual agreement.

(This plan is intentionally unsigned in accordance with dual-anonymous review requirements.)